

Topological Field Arithmetic: Foundations, Obstructions, and a Harmonic-Analysis Reframing

Author: William Paul Sindle **Date:** February 2023 **Status:** Concept paper with a rigorous negative core. Established results are cited; the framework's own claims are labeled as conjectures.

Scope and honesty statement. This paper proposes a reading of "Topological Field Arithmetic" (TFA) that is consistent with known mathematics. It does not claim to prove a new theory. Part 2 proves several *obstructions* (negative results) that constrain what such a theory can be. Part 3 identifies established mathematics — Fourier/Pontryagin duality, generating functions, the circle method, and adelic analysis — in which the core intuition already exists rigorously. The numerical claims about α and ϕ are quarantined in Appendix A and explicitly marked as speculation.

Abstract

The original TFA sketch proposed that numbers be modeled as excitations of a continuous topological medium, that arithmetic operations be reinterpreted as resonance and scattering, and that physical constants emerge as topological thresholds. We show that the first two claims, taken literally, are either vacuous or collapse to ordinary arithmetic (Proposition 2.1), and that the emergent-constant claim is self-undermining once continuous dependence on model parameters is assumed (Theorem 2.4). We then show that the intuitive content of "numbers as waves interacting" is not merely analogous to but *identical with* classical harmonic analysis on locally compact abelian groups (Section 3), and that the most natural "medium" for arithmetic is the adèle ring of the rationals, in the sense of Tate's thesis. The paper concludes with a corrected, minimal formulation of TFA and a list of open problems. The emergence-of-constants proposal is neither supported nor refuted here; it is isolated as an untested speculation.

1. Introduction

The motivating intuition is worth preserving: that arithmetic might be viewed dynamically, through superposition and interference, rather than as a static table of facts. The error of the original sketch was to treat this intuition as *new*. It is not. Additive structure is already represented by convolution and its Fourier dual; arithmetic is already studied through oscillatory integrals; and the topology of the rationals is already the subject of p-adic and adelic analysis. This paper's contribution is to say so

plainly, prove what can be proven against the stronger claims, and propose where genuine research could go.

We do not begin from a clean slate. We begin from a caricature that must be corrected before anything else can be built.

2. Refutations and Repairs

This is the paper's core. Each subsection states a claim, refutes it, and states the repair.

2.1 The representation obstruction: re-description is not replacement

Claim (original). TFA "fundamentally replaces traditional discrete arithmetic" with a field model.

Refutation. Let A be any unital ring and $\varphi : \mathbb{Z} \rightarrow A$ any unital ring homomorphism — that is, any encoding that preserves $0, 1, +, \times$.

Proposition 2.1 (standard). φ is uniquely determined by $\varphi(1) = 1_A$, namely $\varphi(n) = n \cdot 1_A$, and its image is the prime subring:

$$\varphi(\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & \text{char } A = 0, \\ \mathbb{F}_p, & \text{char } A = p. \end{cases}$$

Proof. A unital ring homomorphism preserves 1 and addition, so $\varphi(n) = \varphi(\underbrace{1 + \cdots + 1}_n) = n \cdot 1_A$;

the kernel is an ideal of $\mathbb{Z}$, hence (0) or (p) , and $\mathbb{Z}/\ker \varphi$ is the image. ■

Consequently, *any* structure that correctly reproduces integer arithmetic contains, by construction, an isomorphic copy of arithmetic. The arithmetic identities are inherited from $\mathbb{Z}$, not derived from the medium. The medium can only contribute *extra* structure beyond the image — and that extra structure must be exhibited and justified, not asserted. As stated, "replacement" is unsupported; the stated model is a representation.

Repair. Abandon "replacement." Ask instead: *what new structure does the medium provide, and what does it let us prove that $\mathbb{Z}$ does not?* This turns a false claim into a research question.

2.2 Conservation laws of the proposed form are vacuous

Claim. "Mathematical balance is maintained through conservation laws" replacing symbolic equality.

Refutation. Suppose ν encodes numbers and Φ is any flux functional. The demand $\Phi(u \oplus v) = \Phi(u) + \Phi(v)$ is satisfied *trivially* by $\Phi := \nu^{-1}$, or by any additive functional. Such a law restates $A + B = C$; it does not add dynamics. Any conservation statement has content only

when Φ is defined **geometrically** (independent of the encoding) and is shown to be invariant for non-obvious reasons.

Repair. Specify Φ geometrically and prove invariance as a theorem — or drop the claim.

2.3 "Bounded infinity" conflates cardinality with compactness

Claim. Infinity can be "compressed" into a finite volume by inward fractal folding, creating "infinite complexity in bounded space."

Refutation. This confuses two different notions.

- **Boundedness/compactness** is a metric/topological property. Compact spaces are abundant and can be enormous: $[0, 1]$, the Cantor set, the profinite integers $\hat{\mathbb{Z}}$, and the p-adic integers $\mathbb{Z}_p$ are all compact.
- **Cardinality** is not reduced by self-maps. A function from a set to itself cannot collapse an infinite cardinal to a finite one unless the set was finite. There is no map compressing $2^{\aleph_0}$ into a finite set.

Compactness already delivers "unbounded complexity in a bounded object," for example $\hat{\mathbb{Z}}$ with its profinite topology — but this is a statement about topology, not about shrinking the number of points. The phrase "dimensional compression" is undefined and, read literally, impossible.

Repair. Replace "bounded infinity" with the well-defined question: *can arithmetic be represented by the dynamics of a compact space?* (See Section 3.3; the answer for a large class is yes, via profinite and adelic models.)

2.4 Emergent constants face a genericity problem

Claim. $\alpha \approx 1/137$ and ϕ are topological thresholds of the medium rather than free parameters.

Refutation (partly factual, partly heuristic).

1. **It is not $1/137$.** The measured value is $\alpha \approx 1/137.035999$, and α is scale-dependent (it "runs" under renormalization). "137" is a famously overworked numerical target.
2. **Continuous thresholds are not generically isolated numbers.** A critical value of a smooth functional can typically be moved by smoothly deforming the functional. Isolated, protected values generally arise from *discrete* structure (topological or symmetry protection), which yields integers or rationals — not a transcendental-looking real.
3. Therefore a mechanism that outputs $1/137.035999\dots$ without fitting a parameter would require an integer-like relation or a rigidity theorem. None is supplied.

Theorem 2.4 (restricted no-go for emergent constants). Let $\Theta \subseteq \mathbb{R}^n$ be a connected, open parameter space of models, and let $c : \Theta \rightarrow \mathbb{R}$ be a quantity claimed to "emerge" as a fixed constant

(for example, a critical threshold of a smooth potential). Assume:

(i) c depends continuously on the model $\theta \in \Theta$; and (ii) c is constant on some nonempty open set $U \subseteq \Theta$ (the class of models under consideration).

Then c is constant on all of Θ . Equivalently: a quantity that is both *emergent from a continuously varying model* and *fixed to a specific non-generic value* must in fact be hard-wired — it cannot be a derived output.

Proof. Suppose c is locally constant at some $\theta_0 \in \Theta$. The set $S = \{\theta : c(\theta) = c(\theta_0)\}$ is open by local constancy and closed by continuity, so S is clopen; since Θ is connected and $S \neq \emptyset$, $S = \Theta$. Thus a locally constant c is globally constant.

Conversely, if c is not locally constant at some θ_0 , then by continuity c maps every neighborhood of θ_0 to a set containing an interval of positive length (intermediate value theorem), so c attains every value in that interval for suitable θ . Matching any prescribed measured value $c_\star$ in that interval therefore requires *selecting* θ accordingly — that is, tuning a free parameter. If $c_\star \notin c(\Theta)$ the model simply fails. ■

Corollary 2.5 (topological escape, and why it also fails for α). The only way to evade Theorem 2.4 is for c to be *locally constant while not globally constant* — i.e. a discrete, topologically or symmetry-protected invariant. Such invariants take values in a discrete set (integers, or rationals) and are invariant under continuous deformation. But the measured α is *not* deformation-invariant: it runs with energy scale. Therefore α cannot be a purely topological invariant of a fixed model; and if the model changes discontinuously with scale, assumption (i) fails and the value is an input rather than a derivation.

Remark (what Theorem 2.4 does and does not establish). It is not a proof that no conceivable mechanism fixes a constant. It is a precise statement that, *under continuous dependence*, "emergence" and "fixed to a non-generic value" are mutually exclusive unless the value is discrete. This turns the vague heuristic into a checkable criterion: any claimed derivation must exhibit either (a) an explicit discontinuous/topological invariant, or (b) an explicit tuning parameter, and must say which.

Repair. Move the α, ϕ claim to Appendix A as speculation, and adopt the falsifiable constraint: *any candidate model must output α with zero fitted parameters*. By Theorem 2.4, meeting that constraint forces a discrete invariant, which Corollary 2.5 then rules out for the running α . This is the sharpest available statement of why the original claim fails.

2.5 Summary of refutations

Original claim	Verdict	Repair
"Replaces arithmetic"	Unsupported; representation obstruction (2.1)	Ask what new structure the medium adds
"Conservation replaces equality"	Vacuous as stated (2.2)	Define Φ geometrically; prove invariance
"Bounded infinity / compression"	Ill-posed / impossible as stated (2.3)	Use compact dynamics on $\hat{\mathbb{Z}}, \mathbb{Z}_p$
" α, ϕ emerge"	No-go for continuous models (Thm 2.4); numerically unsupported (2.4)	Zero-parameter test, or remove

3. What Survives: The Intuition Already Exists, Rigorously

The salvageable idea — "numbers as waves that superpose and interfere" — is not merely analogous to known mathematics; in the cases below it *is* known mathematics. TFA should be built on these, not rediscover them by metaphor.

3.1 Addition is convolution; convolution is multiplication (Fourier)

Let G be a locally compact abelian group with dual $\hat{G}$.

Theorem 3.1 (convolution theorem; standard). For $f, g \in L^1(G)$,

$$\widehat{f * g}(\chi) = \hat{f}(\chi) \hat{g}(\chi), \quad \chi \in \hat{G}.$$

Thus **superposition/addition in one representation is multiplication in the dual**. This is the rigorous content of "arithmetic as wave interaction."

Coefficient form. For formal power series,

$$\left(\sum_n a_n x^n \right) \left(\sum_n b_n x^n \right) = \sum_n c_n x^n, \quad c_n = \sum_{i+j=n} a_i b_j,$$

i.e. the **product of generating functions is additive convolution of coefficients**. This is the standard bridge between additive and multiplicative number theory.

3.2 The circle method: additive number theory as oscillatory integrals

The Hardy–Littlewood circle method expresses counts of additive representations (e.g. Goldbach-type and Waring-type problems) as integrals of exponential sums over the circle. This is "resonance and

interference" used as a proof technique, and it is one of the great successes of twentieth-century number theory. Recommending it as TFA's computational engine is both honest and productive.

3.3 The adeles: the true "medium" for arithmetic

The most defensible "topological medium containing the numbers" is not invented — it is the **adele ring** $\mathbb{A}_{\mathbb{Q}}$, a locally compact topological ring in which $\mathbb{Q}$ embeds diagonally, with the p-adic fields $\mathbb{Q}_p$ and the real place completing it. **Tate's thesis** develops arithmetic as harmonic analysis (zetas, L-functions) on $\mathbb{A}$. If TFA wants a field with genuine topology whose arithmetic is a first-class object, this is it.

Related rigorous "number+dynamics" models:

- **Profinite integers / odometers.** $\hat{\mathbb{Z}}$ is a compact group; the odometer (adding machine) is a dynamical system whose orbit structure encodes arithmetic.
- **p-adic analysis.** Ultrametric fields where closeness is number-theoretic.

3.4 Arakelov geometry for "volume" and "conservation"

The original "volume/flux" language has a real home: Arakelov geometry endows arithmetic schemes with metric and volume data, allowing intersection theory and "heights" (a genuine notion of size) on arithmetic objects. This is where conservation-like language can be made precise rather than metaphorical.

4. A Minimal, Corrected Formulation

We can now state TFA without overclaiming.

Definition 4.1. A *field representation of arithmetic* is a topological group or ring M (the medium) together with an injective unital ring homomorphism $\nu : \mathbb{Z} \rightarrow M$ (or $\mathbb{Q} \rightarrow M$) whose image is arithmetic and whose ambient structure satisfies independently motivated conditions.

Definition 4.2. The representation is *non-trivial* if M carries structure not generated by $\nu(\mathbb{Z})$ — e.g. a Haar measure, a dual group, a dynamics — and if that structure yields at least one statement about arithmetic not obtainable from $\mathbb{Z}$ alone.

Program. Find a non-trivial field representation in which a classical theorem is reproved, or a new statement is proved, via the medium's dynamics.

Under this definition, the adelic/p-adic models are field representations, and the circle method is a computational realization. The original sketch is a motivation for Definition 4.1 with no realization attached.

5. Open Problems

1. **Existence/non-existence.** Is there a non-trivial field representation whose dynamics proves a result not accessible by standard methods? (Reproving a known theorem counts as evidence; new theorems count as success.)
 2. **Sharpening Theorem 2.4.** Extend the no-go past continuous dependence: characterize what a "discontinuous emergence" mechanism would have to look like, and quantify the minimum tuning required when (i) fails.
 3. **Classical limit.** For any candidate medium, show how discrete arithmetic arises as a limiting case; absent this, the model does not connect to the theory it claims to generalize.
 4. **Selection principle.** Given the freedom in $\mathbf{P}$ and Φ , find a principle that singles out one model.
 5. **Adelic reformulation.** Can any part of TFA be stated as a theorem about $\mathbb{A}_{\mathbb{Q}}$? If yes, TFA becomes a subfield of existing mathematics rather than a rival to it.
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6. Conclusion

The original TFA text is best read as a *motivation*, not a theory. Its core errors are (i) claiming to replace arithmetic when any faithful model is isomorphic to it (Prop. 2.1), (ii) invoking conservation laws that are vacuous unless geometrically defined (2.2), (iii) an incoherent notion of "bounded infinity" that conflates compactness with cardinality (2.3), and (iv) a no-go for continuously-dependent emergent-constant mechanisms (Theorem 2.4) together with unsupported numerology about α and ϕ (2.4). Its core insight — that addition is convolution and multiplication is its dual — is correct and *already rigorous* in Fourier/Pontryagin analysis, generating functions, the circle method, and adelic arithmetic. The honest path is to build there.

Appendix A — Speculation (Not Supported)

A separate, explicitly speculative appendix. Nothing here is claimed to be true.

One may still ask whether a dimensionless constant could be a topological invariant of some medium. For this to be science it must (a) define the medium and functional concretely, (b) compute a number with **no fitted parameters**, and (c) match $\alpha = 1/137.035999\dots$ exactly. Two cautionary notes: classical topological invariants are discrete (integers/rationals), which makes a transcendental-looking real unlikely from topology alone; and α runs with energy scale, so a purely topological origin is in tension with its observed scale dependence. Treat any claimed derivation with extreme skepticism until (a)–(c) are met. Likewise, the golden ratio appears in *specific* self-similar systems, not in self-similarity in general; it must be derived, not assumed.

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Elementary background used in Theorem 2.4 (intermediate value theorem; connectedness; local constancy) is covered in any standard real-analysis or point-set topology text (e.g. W. Rudin, Principles of Mathematical Analysis; J. R. Munkres, Topology).

Standard results (uniqueness of the complete ordered field; prime subring; convolution theorem) are cited as textbook material and stated without full proof in the text.